Short-Term Urban Traffic Flow Forecasting from Historical Data

Student Research Proposal • Md Omar Hossain • Data Science • Time-Series Analysis • Predictive Modeling

Proposed by	Md Omar Hossain
Project type	Student-led research / prototype study
Estimated duration	8–10 weeks

Project Overview

I want to study whether historical traffic measurements can be used to forecast short-term traffic flow. The project will focus on understanding time-series patterns, building a clear baseline, and comparing a few predictive approaches. The goal is to learn which methods are practical when the available data and computing resources are limited.

Student perspective
I want to keep the first version realistic for a student team: a focused research question, a reproducible experiment, and a prototype that can be discussed with a researcher and improved through collaboration.

Research Problem

Traffic volume changes with time of day, day of week, weather, events, and local conditions. Even a short forecasting horizon can be difficult when the data contains missing values and sudden changes. A controlled student study can compare simple statistical and machine-learning approaches and identify where forecasting becomes unreliable.

Research Objectives

- Explore temporal patterns in a public traffic dataset.
- Prepare a clean and reproducible time-series pipeline.
- Establish a simple historical-average or regression baseline.
- Compare the baseline with one or two machine-learning approaches.
- Analyze forecasting errors during unusual traffic periods.

Research Questions

RQ1. How strong are daily and weekly traffic patterns in the selected dataset?

RQ2. How much improvement does ML provide over a simple baseline?

RQ3. Which time-based features are most useful for short-term forecasting?

RQ4. When does the model produce its largest errors?

Proposed Methodology

Step 1. Select a public time-series traffic dataset with suitable documentation.

Step 2. Perform exploratory analysis of trends, seasonality, missing values, and outliers.

Step 3. Create time-based features and a leakage-safe train/test split.

Step 4. Build a simple baseline and compare it with regression or tree-based models.

Step 5. Evaluate MAE, RMSE, and error distribution across different time periods.

Step 6. Visualize predictions and investigate unusual errors.

Data / Materials / Experimental Resources

Public traffic time-series dataset, Python notebooks, historical timestamps, and optional contextual variables if available.

Evaluation Plan

The main evaluation will use MAE and RMSE on a time-ordered test set. Results will also be broken down by peak and non-peak periods to avoid hiding important differences behind a single average metric.

Expected Challenges

- Missing or irregular observations may reduce forecasting quality.
- Traffic patterns can change over time.
- Random train/test splitting could create data leakage.
- A small dataset may not support complex models reliably.

Reproducibility and Documentation

I will keep the main assumptions, dataset versions, preprocessing decisions, experiment settings, and evaluation results documented. The aim is that another student can understand how the result was produced and repeat the core experiment without relying on undocumented manual steps.

Planned Mitigation

I will begin with a small baseline and a feasibility check. If the original scope becomes too large for the available time or resources, I will narrow the research question rather than adding unnecessary complexity. Limitations and failed experiments will be documented as part of the research process.

Expected Outcomes

- A reproducible traffic forecasting pipeline.
- Baseline-versus-ML comparison.
- Time-based error analysis and visualizations.
- A practical discussion of forecasting limitations.

Potential Research Collaboration

I am looking for someone interested in data science, time-series analysis, or predictive modeling who can contribute to feature engineering, evaluation, or interpretation of the results.

Technical Stack

Python, pandas, NumPy, scikit-learn, matplotlib, Jupyter Notebook.

Scope, Ethics, and Responsible Research

The project will use aggregated or appropriately anonymized public traffic data. No individual driver tracking or personally identifiable information is required.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Data/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, experiment design, implementation, analysis, or writing. I would like to divide responsibilities according to each person's strengths and keep the work transparent and documented.

Possible Next Step

If a researcher is interested, I would first like to discuss feasibility, refine the research question, and agree on a realistic division of work. The proposal can then be updated into a more specific study plan with milestones.

Note: This is a proposed student project. It does not claim completed research or guaranteed results, and the scope can be refined through collaboration.